PFP: Local witnesses and global compatibility in agent composition

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Abstract

Independently checked local answers can disagree on shared variables. We give a finite gluing theorem for explicit assignment tables, retain the overlap disagreements as obstruction evidence, and exhibit a three-agent cycle in which every local task is satisfiable but the joint task is not. The result explains a concrete role for LGGR in PFP agent composition without claiming a general sheaf theorem or automatic multi-agent reliability.

Mechanism inventory

Mechanism	Evidence
Mathematical statement	Written proof under declared hypotheses
Finite example	Standard-library replay of submitted inputs and rejection controls
Model inference	Not executed
Lean verification	No new formalization claimed
Authorization	None supplied by this paper or its replay

1. Local tasks and the global question

Let V be a finite variable set with finite declared domains. Each agent's local task has a scope $S_i \subseteq V$ and an explicitly supplied allowed relation $R_i \subseteq \prod_{v \in S_i} D_v$. A candidate family selects $a_i \in R_i$ for each scope. Local satisfaction and pairwise overlap compatibility are separate checks. Add full-domain unary relations for any uncovered variable when a global-assignment correspondence is required.

2. Finite compatibility theorem

Theorem. A locally permitted family glues to a unique assignment on $\bigcup_i S_i$ precisely when every pair agrees on its intersection. If the scopes cover V, compatible permitted families correspond bijectively to global solutions of all local constraints.

Proof. If a global assignment exists, its restrictions agree. Conversely, for each covered variable choose its value from any containing scope. Pairwise agreement makes that choice independent of the scope. The resulting assignment restricts to each selected row, so it satisfies every relation. Every global assignment with those restrictions has those same values, proving uniqueness. Restriction and gluing are inverse maps when all variables are covered. An uncovered variable with multiple values defeats uniqueness unless its value is supplied separately; an empty uncovered domain makes global existence fail. The full-domain unary relations retain these cases.

3. Three local successes and one global obstruction

Take Boolean variables x, y, z. The first agent checks $x = y$, the second $y = z$, and the third $z \neq x$. Every relation has two rows, so every local existence query succeeds. A global assignment would imply $x = z$ and $x \neq z$, a contradiction. In the selected family $(x, y) = (0, 0)$, $(y, z) = (0, 0)$, $(z, x) = (0, 1)$, the overlap value of x disagrees. Replacing the final relation by equality gives the two global solutions $(0, 0, 0)$ and $(1, 1, 1)$ and compatible local restrictions for each.

This is the finite LGGR distinction: local green results need not produce a global green result. A disagreement is an obstruction to the declared gluing requirement, not an assertion that all differences are inherently bad.

4. Agent packets and the integration check

The shared prompt explains row selection and the possibility of overlap failure. The typed instance supplies domains, scopes, tables, and the joint question. Each agent returns its actual row and local certificate. The integrator replays local membership and checks common-variable values before admitting a global witness. A loss ledger or a narrative agreement cannot replace these values. Renaming variables requires explicit bijections applied to the task and every witness; dropping an overlap changes the problem.

5. Executable falsifiers

The reference replay independently enumerates all eight original global assignments and checks submitted local rows. It verifies local nonemptiness of the incompatible cycle, refuses its disagreeing family, and accepts the compatible equality family. A global solution is decoded only from a family passing the overlap checks. There are no model calls, observed swarm improvements, or new claims about physical contextuality.

6. Scope and relation to WDP

The assignment gluing fact is standard. Its PFP use is a task-bound composition contract that preserves local evidence while exposing missing global compatibility. WDP §20.7 supplies the broader finite constraint construction and zero-disagreement encoding. More general agent composition additionally requires contracts for effects, authority, timing, and representation change. Those cannot be established from this static Boolean example alone.

The two-level teaching packet includes shared system rules, this typed user instance, a different worked example and a required candidate response schema. It keeps provider system/user roles separate. It is a saved example, not a schema-admission or model-run receipt; the local replay checks the mathematical or protocol obligations stated above.

7. Typed operation composition is a different obligation

The current OEOP compiler checks ordered input and output slots, declared effects and source-bound operation identities. This is not the assignment gluing theorem: two callbacks with compatible types can still disagree on task identity, candidate bytes or shared state. The execution boundary must retain those bindings and observe the actual operations. The finite theorem cannot supply an unimplemented callback, undo a side effect or resolve a missing native host.

In the bounded DDC repair study, the operation order is C, R, E, I, M, T. Every stage carries the same run key, request digest and candidate identity. Compilation uses the existing typed workflow and operation-inventory boundaries; Woland owns the working copy and fixed checks. OEEIS screening and Prompt Civilization lineage are injected services, not responsibilities absorbed into the capability catalog.

A complete six-step execution is a concrete integration observation, not a general gluing theorem for effectful agents. The example's original source is unchanged, activation remains on hold, and no mathematical residual is measured. The assignment proof and Boolean falsifiers above are unchanged; they remain independent from this implementation study.

Sources and relation to the programme

This draft builds on the WDP manuscript, Paper 021's source-and-execution distinctions, and the corrected findings notices. Paper 022 supplies a research design whose live-model and offline-strategy claims require their own source-bound scrutiny. These drafts do not adopt its empirical tables as new measurements. The PFP collection provides the `shared-system/typed-instance` protocol. Genesis and MSS-N presentation stages do not license a new mathematical or physical carrier.